

Bahram Afsharmanesh

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RESEARCH INTERESTS

High-performance computing and machine learning systems; heterogeneous GPU scheduling and distributed training; GPU kernels and collective communication; communication-computation overlap, quantization, and performance modeling for large language model inference.

EDUCATION

University of Georgia	2023–Present
Ph.D. in Computer Science, in progress	GPA: 3.95/4.00
Advisor: Prof. Gagan Agrawal.	
Sharif University of Technology	2012–2014
M.Sc. in Mechanical Engineering	GPA: 3.81/4.00
Iran University of Science and Technology	2008–2012
B.Sc. in Mechanical Engineering	GPA: 3.76/4.00

SELECTED PUBLICATIONS

- [1] **Bahram Afsharmanesh**, Md Musfiqur Rahman Sanim, AmirAli Mirian, and Gagan Agrawal. **HeteroSched: Co-Optimizing Scheduling and Parallelization for Deep Learning Workloads for Heterogeneous GPU Clusters**. *International Conference on Parallel Architectures and Compilation Techniques (PACT)*, 2026. Accepted.
- [2] Md Musfiqur Rahman Sanim, Zhihao Shu, **Bahram Afsharmanesh**, AmirAli Mirian, Wei Niu, and Gagan Agrawal. **Mask-Aware Execution for Efficient JEPA Training**. *International Conference on Parallel Architectures and Compilation Techniques (PACT)*, 2026. Accepted.
- [3] AmirAli Mirian, **Bahram Afsharmanesh**, Md Musfiqur Rahman Sanim, Wei Niu, and Gagan Agrawal. **Fast Sparse Linear Algebra for LLMs with Contextual Sparsity**. *IEEE International Conference on High Performance Computing, Data, and Analytics (HiPC)*. Submitted.
- [4] Md Musfiqur Rahman Sanim, Zhihao Shu, **Bahram Afsharmanesh**, AmirAli Mirian, Jiexiong Guan, Wei Niu, Bin Ren, and Gagan Agrawal. **Optimizing 3D Gaussian Splattering for Mobile GPUs**. *International Conference on Parallel Architectures and Compilation Techniques (PACT)*, 2025. [Preprint]

PROFESSIONAL SERVICE

Artifact Evaluation Committee Member	2026
International Conference on Parallel Architectures and Compilation Techniques (PACT).	

RESEARCH EXPERIENCE

Graduate Research Assistant

Aug. 2023–Present

University of Georgia; advisor: Prof. Gagan Agrawal

- Designed HeteroSched to jointly optimize scheduling and parallelization for heterogeneous GPU clusters; evaluated with performance simulation and physical experiments.
- Developed CUDA/C++ and MSCCL++ collective-communication kernels, fine-grained communication-computation overlap, and quantized AllReduce for distributed LLM inference over PCIe and NVLink.
- Built microbenchmarks and analytical performance models; used Nsight Systems and Nsight Compute to study synchronization, memory bandwidth, kernel occupancy, and end-to-end performance.

Research Assistant

May 2020–June 2021

Amirkabir University of Technology

- Investigated machine learning as a function approximator for numerical PDE solvers; developed MATLAB applications for mechanical-system analysis under static, dynamic, and thermal loads.

Research Assistant

Sept. 2012–Sept. 2015

Sharif University of Technology

- Developed PDE solvers in MATLAB and molecular dynamics algorithms in Fortran; used genetic algorithms to calibrate numerical models against molecular dynamics simulations.

Undergraduate Researcher

Sept. 2010–Sept. 2012

Iran University of Science and Technology; advisor: Prof. S. M. Hasheminejad

- Solved PDEs for laminated plates using MATLAB and Mathematica.

INDUSTRY EXPERIENCE

Data Scientist

Aug. 2021–Dec. 2022

Baroline

- Built spam-detection and named-entity-recognition models with PyTorch, BERT, and scikit-learn; deployed inference APIs using FastAPI, Docker, and Kafka, reducing response latency by 25%.
- Developed data pipelines and warehouses with Airflow and PostgreSQL, and analytics dashboards with Pandas, Streamlit, Power BI, and Metabase.

R&D Engineer

Jan. 2016–Apr. 2020

MAPNA PARS Company

- Built MATLAB toolboxes for system identification and eigenvalue analysis; developed regression models to predict component behavior and accelerate engineering design workflows.

SELECTED TECHNICAL PROJECTS

CUDA Parallel Prefix Scan. Implemented a hierarchical GPU scan using shared memory, custom kernels, multilevel residue computation, and uniform addition for large inputs.

Deep Neural Network Pruning. Implemented one-shot and iterative magnitude pruning, including unstructured and filter pruning; evaluated sparsity and model-performance tradeoffs.

TECHNICAL SKILLS

Programming	C, C++, Python, CUDA, Fortran, SQL, MATLAB, Mathematica, Bash
GPU and parallel	MSCCL++, NCCL, CUTLASS/CuTe, Triton, MPI, OpenMP
ML and data	PyTorch, JAX, scikit-learn, NumPy, Pandas, FastAPI, Kafka, Airflow
Performance	Nsight Systems, Nsight Compute, profiling, benchmarking, analytical performance modeling, roofline analysis
Development	Linux, Git, Docker, Make/CMake, CI/CD, AWS, L ^A T _E X